PYTHON FUNDAMENTALS

# Lesson 1.1: Variable Declaration & Naming Conventions

Chapter 1: Variables and Primitive Data Types

## LEARNING OBJECTIVES

- Understand Python's object reference model and dynamic typing.
- Use the syntax for assignment, reassignment and multiple assignment.
- Apply the PEP 8 naming conventions.
- Identify, fix and prevent common variable initialization and naming errors.

CORE CONCEPTS

# Variables & Dynamic Memory Allocation

## What is a Variable in Python?

- A variable is a name (label) that refers to an object stored in memory, not a fixed memory box.
- It is created automatically the first time you assign a value with the assignment operator (=).
- The name holds a reference to the object; the object holds the value and its type.

## Dynamic Typing Mechanism

- **Types belong to objects, not variables:** no type declaration is needed before assignment.
- **Run-time typing:** the same name can later refer to an object of a different type.
- **Garbage collection:** objects that are no longer referenced are reclaimed automatically.

## Memory Reference Inspection

```
# The name 'score' refers to an int object
score = 100
print(type(score), id(score))
# <class 'int'> 4370193552   (id varies)

# Rebinding 'score' to a new str object
score = "A+ Distinction"
print(type(score), id(score))
# <class 'str'> 4373841904   (id varies)
```

**Key takeaway:** reassignment does not change the old object. It binds the name to a different object; an object that nothing refers to any more can be reclaimed.

SYNTAX & EXAMPLES

# Variable Declaration & Assignment Syntax

## Fundamental Syntax: `variable_name = expression`

- **Left-hand side:** a valid identifier.
- **Assignment operator (=):** binds the name to the object produced by the right-hand side.
- **Right-hand side:** evaluated completely before the name is bound.

## Supported Assignment Patterns

- **Standard assignment:** one name bound to the value of one expression.
- **Chained assignment:** several names bound to the same object. This is safe for immutable values; page 5 shows the pitfall with a mutable list.
- **Tuple unpacking:** several names assigned in one statement.
- **Swapping:** values exchanged without a temporary variable.

## Code Demonstrations

```
# 1. Standard single assignment
course_name = "Python Fundamentals"
credit_hours = 3
passing_gpa = 2.5
is_published = True

# 2. Chained assignment (shared reference)
min_score = default_score = 0

# 3. Tuple unpacking / multiple assignment
x_coord, y_coord, z_coord = 10.5, -4.2, 0.0

# 4. Swapping two values
left_node, right_node = "Alpha", "Beta"
left_node, right_node = right_node, left_node
```

STANDARDS & CONVENTIONS

# Identifier Rules & PEP 8 Naming Conventions

## Mandatory Interpreter Rules (Syntax)

- **Allowed characters:** letters (a–z, A–Z), digits (0–9) and underscores (_).
- **Leading character:** cannot be a digit (`1st_place` is invalid).
- **Case sensitivity:** `user_id`, `User_Id` and `USER_ID` are three distinct names.
- **No reserved keywords:** Python's 35 keywords (`if`, `for`, `class`, `def`, `return`, `None`, `global`, ...) cannot be names.

## PEP 8 Style Guidelines (Best Practices)

- Use `snake_case` for variables and function names.
- Use `UPPER_SNAKE_CASE` for module-level constants.
- Prefer descriptive, intention-revealing names to single-letter abbreviations.

## Naming Comparison Matrix

| Category | Valid & Pythonic | Invalid / Discouraged |
|---|---|---|
| Standard variable | `student_count` | `studentCount`<br>camelCase |
| Boolean flag | `is_authenticated` | `flag1`<br>vague meaning |
| Constant | `MAX_RETRY_LIMIT` | `max-retry`<br>hyphen: syntax error |
| Internal / private | `_cache_buffer` | `2nd_buffer`<br>starts with a digit |
| Reserved word | `class_room_id` | `class`<br>SyntaxError |

**Tip:** run `import keyword; print(keyword.kwlist)` to list all 35 reserved keywords.

TROUBLESHOOTING & PITFALLS

# Common Variable Mistakes & How to Fix Them

| Error type | Problematic code | Corrected code and resolution |
| --- | --- | --- |
| **1. Uninitialized variable**<br>NameError | `print(total_price)`<br>The name is read before it is assigned, or it is misspelled. | `total_price = 0.0`<br>`print(total_price)`<br>Assign a value before reading a name. |
| **2. Shadowing a built-in**<br>TypeError later | `list = [10, 20, 30]`<br>`new_list = list("abc")`<br>The name `list` now refers to a list, not the built-in constructor. | `scores_list = [10, 20, 30]`<br>`chars = list("abc")`<br>Avoid names such as list, str, int, id, sum and max. |
| **3. Unpacking mismatch**<br>ValueError | `a, b = 10, 20, 30`<br>Too many values to unpack: two names, three values. | `a, b, c = 10, 20, 30` or `a, *b = 10, 20, 30`<br>Match the number of names, or use star unpacking. |
| **4. Shared mutable reference**<br>Logic bug | `row_a = row_b = []`<br>`row_a.append(1)`<br>Both names refer to the same list, so row_b is also [1]. | `row_a, row_b = [], []`<br>`row_a.append(1)`<br>Create a separate mutable object for each name. |

KNOWLEDGE CHECK

# Mini Practice & Concept Checkpoint

**Q1 (multiple choice): Which variable name is BOTH syntactically valid in Python and compliant with PEP 8?**

**A)** `2nd_user_score`

**B)** `user-profile-id`

**C)** `max_connection_timeout`

**D)** `global`

KNOWLEDGE CHECK · ANSWER

# Mini Practice & Concept Checkpoint

**Q1 (multiple choice): Which variable name is BOTH syntactically valid in Python and compliant with PEP 8?**

**A)** `2nd_user_score`

**B)** `user-profile-id`

**C)** `max_connection_timeout`

**D)** `global`

**Answer: C.** It uses only lowercase letters and underscores (snake_case). A starts with a digit, B contains hyphens and D is a reserved keyword.

KNOWLEDGE CHECK

# Mini Practice & Concept Checkpoint

**Q2: What is printed after this snippet runs?**

```
x = 15; y = x; x = 40; print(y)
```

A) 40

B) 15

C) None

D) NameError

# Mini Practice & Concept Checkpoint

**Q2: What is printed after this snippet runs?**

```
x = 15; y = x; x = 40; print(y)
```

A) 40

B) 15

C) None

D) NameError

**Answer: B.** y = x binds y to the object 15. Rebinding x to 40 afterwards does not change what y refers to.

KNOWLEDGE CHECK

# Mini Practice & Concept Checkpoint

## Buggy Code Snippet

```
# Fix 3 issues in this code:
1st_student = "Alice"
class = "PY101"
StudentAge = 20
```

# Mini Practice & Concept Checkpoint

## Suggested Fix

```
first_student = "Alice"   # a name cannot start with a digit
class_code = "PY101"      # class is a reserved keyword
student_age = 20          # PEP 8: snake_case for variables
```

The first two lines are syntax errors and stop the program from running. The third line runs, but it does not follow the PEP 8 naming convention.

SUMMARY

# The End

## Thank you

### Primary Technical References


- Van Rossum, G., Warsaw, B., & Coghlan, N. (2001). *PEP 8 – Style Guide for Python Code*. Python Software Foundation. https://peps.python.org/pep-0008/
- Python Software Foundation. *The Python Language Reference*: Execution Model and Data Model. https://docs.python.org/3/reference/


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